2. Finite Dim Vector Spaces

- linear combination $a_1 v_1 + \cdots + a_m v_m$

  where $a_1, \ldots, a_m \in \mathbb{F}$ and $v_1, \ldots, v_m \in V$

- $\text{span}(v_1, \ldots, v_m) = \{\underbrace{a_1 v_1 + \cdots + a_m v_m}_{\text{linear combination}} : a_1, \ldots, a_m \in \mathbb{F}\}$

  is the smallest subspace containing $v_1, \ldots, v_m$

- $\text{span}() = \{0\}$

$\ulcorner$ $\text{span}(v_1, \ldots, v_m) = \{a_1 v_1\} + \cdots + \{a_m v_m\}$

  is smallest subspace containing $\underbrace{\{a_1 v_1\}}_{\text{closed under mul}}, \ldots, \{a_m v_m\}$

- $\begin{bmatrix} 2 & 1 \\ 1 & -2 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 6 \\ 5 \end{bmatrix} = \begin{bmatrix} 17 \\ -4 \\ 2 \end{bmatrix} \in \text{span}(\bullet, \bullet)$

- $\begin{bmatrix} 17 \\ -4 \\ 5 \end{bmatrix} \notin \text{span}(\bullet, \bullet)$ i.e not linear combination.

- $\underbrace{\text{list } v_1, \dots, v_m}_{\text{list has finite length}}$ spans V if $\operatorname{span}(v_1, \dots, v_m) = \underbrace{V}_{\text{"finite dimensional"}}$

e.g. $(1, \dots, 0)$ $\dots$ $(0, \dots, 1)$ spans $\mathbb{F}^n = \{(x_1, \dots, x_n)\}$

- polynomial $p : \mathbb{F} \to \mathbb{F}$: $p(z) = a_0 + a_1 z + \dots + \underbrace{a_m z^m}_{\text{if } a_m \neq 0 \text{ then } \deg p = m}$

- $\deg 0 = -\infty$

- $\mathcal{P}(\mathbb{F})$: polynomials with ① coefficients in $\mathbb{F}$

- $\mathcal{P}_m(\mathbb{F})$: polynomials with ① coefficients in $\mathbb{F}$
  ② $\deg \leq m$

  m ↓ non-negative int

---

vector space $\begin{cases} \text{finite-dimensional} & \text{e.g. } \mathcal{P}_m(\mathbb{F}) = \operatorname{span}(1, z, \dots, z^m) \\ \text{infinite-dimensional} & \text{e.g. } \mathcal{P}(\mathbb{F}) \end{cases}$

$\because$ $\forall m \in \mathbb{N}_0$ then $\operatorname{span}(1, \dots, z^m) \subsetneq \mathcal{P}(\mathbb{F})$ (like z^{m+1})

- list $v_1, \dots, v_m$ is linearly independent

if $a_1 v_1 + \dots + a_m v_m = 0$ requires $a_1 = \dots = a_m = 0$

linear combination

$\Longleftrightarrow$ unique expression

$$v = a_1 v_1 + \dots + a_m v_m \quad \text{for } \forall v \in \operatorname{span}(v_1, \dots, v_m)$$

- empty list $()$ is linearly independent.

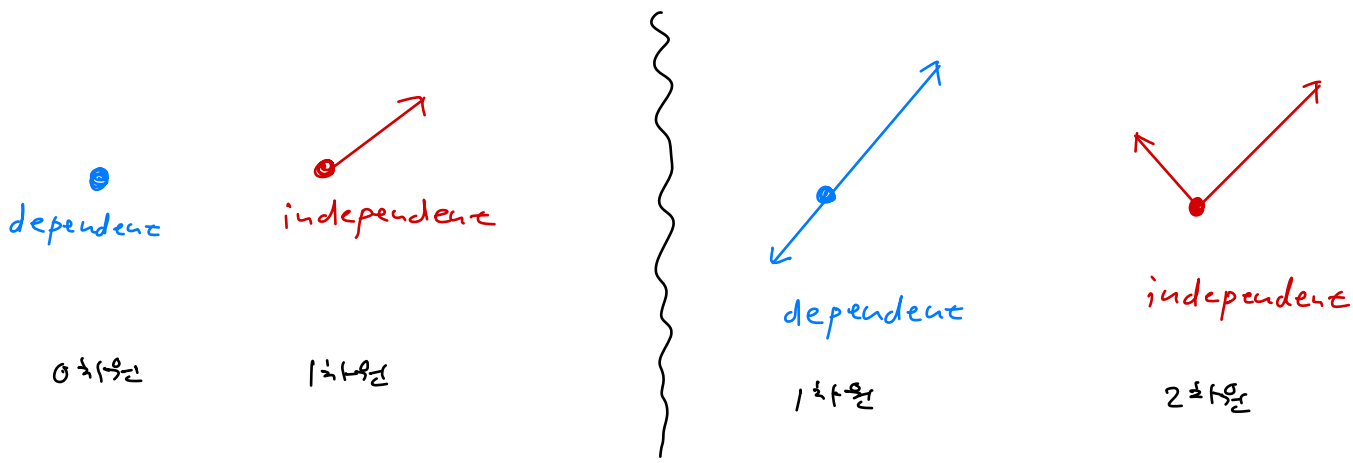
---

list $1, z, \dots, z^m$ linearly independent

$\because$

$$a_0 \cdot 1 + a_1 z_1 + \dots + a_m z^m = 0 \quad (\forall z \in \mathbb{F})$$

requires $(a_0, a_1, \dots, a_m) = (0, 0, \dots, 0)$



- list of vectors $\begin{cases} \text{linearly dependent} \\ \text{linearly independent} \end{cases}$

$v_1, \dots, v_m$ iff $a_1 v_1 + \dots + a_m v_m = 0$

linearly dependent need not to be $a_1 = \dots = a_m = 0$

- suppose $v_1, \dots, v_m$ linearly dependent

① $v_k \in \operatorname{span}(v_1, \dots, v_{k-1})$ $(\exists k$ in $1, \dots, m)$

② $\operatorname{span}(v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_m) = \operatorname{span}(v_1, \dots, v_m)$

$\because$
- $a_1 v_1 + \dots + a_m v_m = 0$
- select $k = \max\{k : a_k \neq 0\}$
- replace v_k with $b_1 v_1 + \dots + b_{k-1} v_{k-1}$

---

independent list is no longer than spanning list.

$v_1, \dots, v_m$ $\qquad$ $w_1, \dots, w_n$

iteration ① spanning list $\qquad$ independent $v_1, \dots, v_{k-1}\ w_k \dots w_n$

② linearly dependent list $\qquad$ independent $v_1, \dots, v_{k-1}\ v_k\ \cancel{w_k} \dots w_n$

$\Rightarrow$ one of w's is linear combination of prev vectors

---

finite dimensional vector space $\Rightarrow$ subspace is finite dimensional

$\because$
- repeatedly add $u_k \notin \operatorname{span}(u_1, \dots, u_{k-1})$
- $u_1, \dots, u_k$ is linearly independent
- linearly independent list is no longer than spanning list of V $\Rightarrow$ terminates

2B Bases

- basis of V is a list of vectors that is
  1. linearly independent
  2. spans V

$\Longleftrightarrow$ unique expression $v = a_1 v_1 + \cdots + a_n v_n$ for $\forall v \in V$ (basis)

$\Longleftrightarrow$ $\underbrace{\{a_1 v_1\}}_{\text{line}} \oplus \cdots \oplus \underbrace{\{a_n v_n\}}_{\text{line}} = V$

$\Rightarrow$ ($\Rightarrow$) let $v = a_1 v_1 + \cdots + a_n v_n$
$= c_1 v_1 + \cdots + c_n v_n$

then $0 = \underbrace{(a_1 - c_1)}_{=0} v_1 + \cdots + \underbrace{(a_n - c_n)}_{=0} v_n$

($\Leftarrow$) ① (independent) unique $0 = \underbrace{a_1}_{=0} v_1 + \cdots + \underbrace{a_n}_{=0} v_n$

② (span) every $v \in V$ ... blah blah blah

- $\mathbb{F}^n$ $\rightarrow$ $(1, 0, \ldots, 0), (0, 1, \ldots, 0), \ldots, (0, 0, \ldots, 1)$
- $\{(x, x, y)\}$ $\rightarrow$ $(1, 1, 0), (0, 0, 1)$
- $\{(x, y, z) : x + y + z = 0\}$ $\rightarrow$ $(1, -1, 0), (1, 0, -1)$
- $\mathcal{P}_m(\mathbb{F})$ $\rightarrow$ $1, z, \ldots, z^m$

spanning list can be reduced to basis

$\because$

independent

if $v_k \in \operatorname{span}(v_1, \ldots, v_{k-1}) \longrightarrow$ remove v_k

---

- finite-dim vector space has basis

  because it has spanning list to be reduced

---

linearly independent list of vectors in finite-dim

vector space can be extended to basis

$\because$ $\underbrace{u_1, \ldots, u_m}_{\text{independent}}$, $\underbrace{w_1, \ldots, w_n}_{\text{spanning list}}$ and reduce it step-by-step.

---

if U is subspace of finite-dim V

then W exists with $U \oplus W = V$

$\because$

- basis of $U = \{u_1, \ldots, u_m\}$
- basis of $V = \{u_1, \ldots, u_m, w_1, \ldots, w_n\}$
- basis of $W = \{w_1, \ldots, w_n\}$

2C. Dimension

---

any two basis of

finite-dim vector space has same length

$\because$ independent list no longer than spanning list

- dimension of finite-dim vector space

  is the length of any basis of vector space

- $\dim \mathbb{F}^n = n$
- $\dim \mathcal{P}_m(\mathbb{F}) = m+1$

- $\dim U \leq \dim V$ if ① U is subspace of V

  ② V is finite-dim.

$\because$ basis of U : independent list

basis of V : spanning list.

- independent list of vectors

  with length $= \dim V$ is basis of V (V finite-dim)

$\ulcorner$ reduce $\underbrace{v_1, \ldots, v_n}_{\text{independent}}$, $\underbrace{\cancel{w_1}, \ldots, \cancel{w_m}}_{\text{spanning list}}$

but basis should be length $= n$

- if $\dim U = \dim V$ (V finite-dim)

  with U subspace of V then $U = V$

- $U = \{ p \in \mathcal{P}_3(\mathbb{R}) : p'(5) = 0 \}$

- 1, $(x-5)^2$, $(x-5)^3$ are linearly independent.

  considering $a \cdot 1 + b(x-5)^2 + c(x-5)^3 = 0$

- $3 \leq \dim U \leq \dim \mathcal{P}_3(\mathbb{R}) = 4$

- $\dim U \neq 4$ otherwise $U = \mathcal{P}_3(\mathbb{R})$

  but $x \notin U$

- $\dim U = 3$ therefore 1, $(x-5)^2$, $(x-5)^3$ are basis.

- spanning list of vectors

  with length $= \dim V$ is basis of V (V finite-dim)

$\because$ reduce $v_1, \dots, v_n$ to basis

but basis should be length $= n$

$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2)$

if V_1, V_2 is subspace of finite-dim vector space

---

basis of $V_1 \cap V_2$: $v_1, \dots, v_n$

basis of V_1 : $v_1, \dots, v_n, u_1, \dots, u_j$

basis of V_2 : $v_1, \dots, v_n, w_1, \dots, w_k$

Claim $v_1, \dots, v_n, u_1, \dots, u_j, w_1, \dots, w_k$

is linearly independent

$\because$ $a_1 v_1 + \dots + a_n v_n$

$+ b_1 u_1 + \dots + b_j u_j$

$+ c_1 w_1 + \dots + c_k w_k = 0$

then $\underset{\substack{\| \\ 0}}{c_1 w_1} + \dots + \underset{\substack{\| \\ 0}}{c_k w_k} = \underset{\substack{\| \\ 0}}{d_1 v_1} + \dots + \underset{\substack{\| \\ 0}}{d_m v_m}$

- linear combination $a_1 v_1 + \cdots + a_m v_m$

- $\operatorname{span}(v_1, \ldots, v_m) = \{a_1 v_1 + \cdots + a_m v_m : a_1, \ldots, a_m \in \mathbb{F}\}$

  $= \{a_1 v_1\} + \cdots + \{a_m v_m\}$

  is smallest subspace containing $v_1, \ldots, v_m$

- $v_1, \ldots, v_m$ linearly independent

  $\Longleftrightarrow$ $\operatorname{span}(v_1, \ldots, v_m) = \{a_1 v_1\} \oplus \cdots \oplus \{a_m v_m\}$

  $\Longleftrightarrow$ unique expression $\underbrace{v}_{\substack{\text{sufficient} \\ \text{to check only } 0}} = a_1 v_1 + \cdots + a_m v_m$ for $v \in \operatorname{span}(v_1, \ldots, v_m)$

- V is ~~finite-dim~~ if

  there is a list that spans V

- polynomial $\underset{\substack{\text{infinite} \\ \text{-dim}}}{\mathcal{P}(\mathbb{F})}$, $\underset{\substack{\text{finite} \\ \text{-dim}}}{\mathcal{P}_m(\mathbb{F})}$

- $1, z, \ldots, z^m$ basis of $\mathcal{P}_m(\mathbb{F})$ i.e ① spanning list

  ② linearly independent.

---

linearly dependent $v_1, \ldots, v_m$

k s.t ① $v_k = \operatorname{span}(v_1, \ldots, v_{k-1})$

② $\operatorname{span}(v_1, \ldots, v_{k-1}, v_{k-1}, \ldots, v_m)$

$= \operatorname{span}(v_1, \ldots, v_m)$

Note.

$a_1 v_1 + \cdots + a_m v_m = 0$

$k = \max\{i : a_i \neq 0\}$

independent list is no longer than spanning list.

$v_1, \ldots, v_m$ $\qquad$ $w_1, \ldots, w_n$

iteration ① spanning list $\qquad$ independent $v_1, \ldots, v_{k-1}, w_k, \ldots, w_n$

② linearly dependent list $\qquad$ independent $v_1, \ldots, v_{k-1}, v_k, \cancel{w_k}, \ldots, w_n$

$\Rightarrow$ one of w's is linear combination of prev vectors

---

finite-dim space V $\Rightarrow$ subspace U is finite-dim

$\because$

- exhaust all $a_k \notin \operatorname{span}(a_1, \ldots, a_{k-1})$ from U $\qquad$ independent

- inside V, independent list is no longer than spanning list